

Hamna Raeel

Software Engineer, ML/AI

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Summary

Machine Learning Engineer with hands-on experience training deep-learning segmentation models for medical imaging and implementing end-to-end data pipelines on multi-cloud platforms. Proven ability to develop and optimize PyTorch/TensorFlow models, containerize and orchestrate workloads with Docker/Kubernetes, and automate deployments via Jenkins CI/CD. Familiar with monitoring and observability using Prometheus and Grafana, and experienced in deploying applications on IBM Cloud and Azure.

Skills

Programming: Python

Machine Learning: PyTorch, TensorFlow

Cloud/MLOps: Docker, Kubernetes, Jenkins CI/CD, Prometheus, Grafana, IBM Cloud, Azure

Experience

Research Assistant, Artificial Intelligence Diagnostic Lab, FAST-NUCES – Islamabad Oct 2025 – Jan 2026

- Trained deep learning segmentation models (Mask & Fast R-CNN, SegFormer, DETR) for histopathology analysis.
- Optimized preprocessing, augmentation, and training pipelines to boost model performance and efficiency.
- Built and evaluated classification models (ResNet, DenseNet, EfficientNet, Vision and Swin Transformer).
- Improved model performance through systematic feature extraction, hyperparameter optimization, and evaluation.

DevOps and Cloud Infrastructure Intern, Software Productivity Strategists, Inc. (SPS) – Islamabad Jul 2025 – Sep 2025

- Developed ELT data pipelines and automated CI/CD deployment workflows using Jenkins for reliable model releases.
- Containerized applications with Docker and orchestrated deployments using Kubernetes on IBM Cloud.
- Implemented monitoring and logging using Prometheus and Grafana for cloud-native applications.
- Managed deployments and infrastructure across multi-cloud environments, including Azure, IBM Cloud, and VMware.

Data Science Intern, Artificial Intelligence Diagnostic Lab, FAST-NUCES – Islamabad Mar 2025 – May 2025

- Implemented RAG pipelines using vector databases, embedding models, and LLMs for document Q&A.
- Developed and evaluated ML/DL models (CNN, ResNet, UNet, SVM, Decision Trees) across benchmark datasets.
- Applied ML to medical imaging and retail datasets, including CBIS-DDSM, Chest X-Ray, Retinal Vessel, and CIFAR-10/100.

Software Engineer, Optimus Bee – Lahore Nov 2022 – Dec 2024

- Developed a Mental Health Wellness application with an AI-powered chatbot, mood tracking, and recommendations.
- Built an e-commerce platform with integrated Stripe payment processing, authentication, and order management.
- Created a Financial Budget Management Tool for expense tracking, categorization, and analytics dashboards.
- Built an admin dashboard with dynamic UI, REST APIs, and reporting for an Australian travel agency.

Front End Developer, Cyber Legends – Lahore Jun 2021 – Oct 2022

- Led development of a WordPress-based university portal (NASTP-IIT) with a custom online admission system.

- Built "Infinite Writing," a freelance services platform with dual client/writer dashboards using HTML, Tailwind CSS, and JavaScript.

Projects

Research & ML

Segmentation-Guided Histopathology Analysis for Hirschsprung Disease Detection (*Python, PyTorch*)

- 92% Dice Score on ganglion cell segmentation 92%

Meta-Learning for Zero-Shot Cross-Lingual Transfer (CML-mDeBERTa) (*Python*)

- 80.34% mean accuracy 80.34%

Engineering & Full-Stack

CloudETL – ELT pipelines with CI/CD, Docker, and Kubernetes on IBM Cloud (*Docker, Kubernetes, Jenkins CI/CD*)

WebApp – Multi-cloud deployment with monitoring (*Azure, IBM Cloud, VMware, Prometheus, Grafana*)

Mental Health Wellness App – AI-powered chatbot and tracking (*React, Node.js*)

Financial Budget Management Tool – Expense analytics (*React, Node.js*)

Travel Agency Admin Dashboard – Dynamic UI with REST APIs (*React, Node.js, REST APIs*)

Education

Master of Science, Data Science, FAST, National University of Computer and Emerging Sciences (NUCES) Jan 2023

Certifications & Trainings

- AI Fluency: Framework & Foundations, **Anthropic**
- Google Advanced Data Analytics, **Google**
- AI Concepts for Developers, **Microsoft**
- AI Apps & Agents on Azure, **Microsoft**
- ChatGPT Prompt-Engineering for Developers, **DeepLearning.AI**
- Databases and SQL for Data Science with Python, **IBM**